16. ROS2 Common Command Tools

1. Package Management Tool ros2 pkg

1.1. ros2 pkg create

Function: create a function package. When creating one, you need to specify the package name, the build type, the dependencies, etc.

Format:

```
ros2 pkg create <package_name> --build-type <build-type> --dependencies <dependencies>
```

In the ros2 command:

- **pkg**: indicates the functions related to function packages;
- **create**: indicates the creation of a function package;
- **package_name**: required: the name of the new function package;
- **build-type**: required: indicates whether the newly created function package is C++ or Python. If C++ or C is used, follow ament_cmake; if Python is used, follow ament_python;
- **dependencies**: optional: indicates the dependencies of the function package. A C++ function package needs to include rclcpp, and a Python function package needs to include rclpy, as well as any other required dependencies;

1.2. ros2 pkg list

Function: view the list of function packages in the system

Format:

```
ros2 pkg list
```

```
yahboom@yahboom-virtual-machine:~$ ros2 pkg list
action_msgs
action_tutorials_cpp
action_tutorials_interfaces
action_tutorials_py
actionlib_msgs
ament_cmake
ament_cmake_auto
ament_cmake_copyright
ament_cmake_core
ament_cmake_cppcheck
ament_cmake_cpplint
ament_cmake_export_definitions
ament_cmake_export_dependencies
ament_cmake_export_include_directories
ament_cmake_export_interfaces
ament_cmake_export_libraries
ament_cmake_export_link_flags
ament_cmake_export_targets
ament_cmake_flake8
ament_cmake_gmock
ament_cmake_gtest
ament_cmake_include_directories
ament_cmake_libraries
ament_cmake_lint_cmake
ament_cmake_pep257
ament_cmake_pytest
ament_cmake_python
ament_cmake_ros
ament_cmake_target_dependencies
```

1.3. ros2 pkg executables

Function: view all executable files within a certain package

Format:

```
ros2 pkg executables pkg_name
```

```
yahboom@yahboom-virtual-machine:~$ ros2 pkg executables turtlesim
turtlesim draw_square
turtlesim mimic
turtlesim turtle_teleop_key
turtlesim turtlesim_node
```

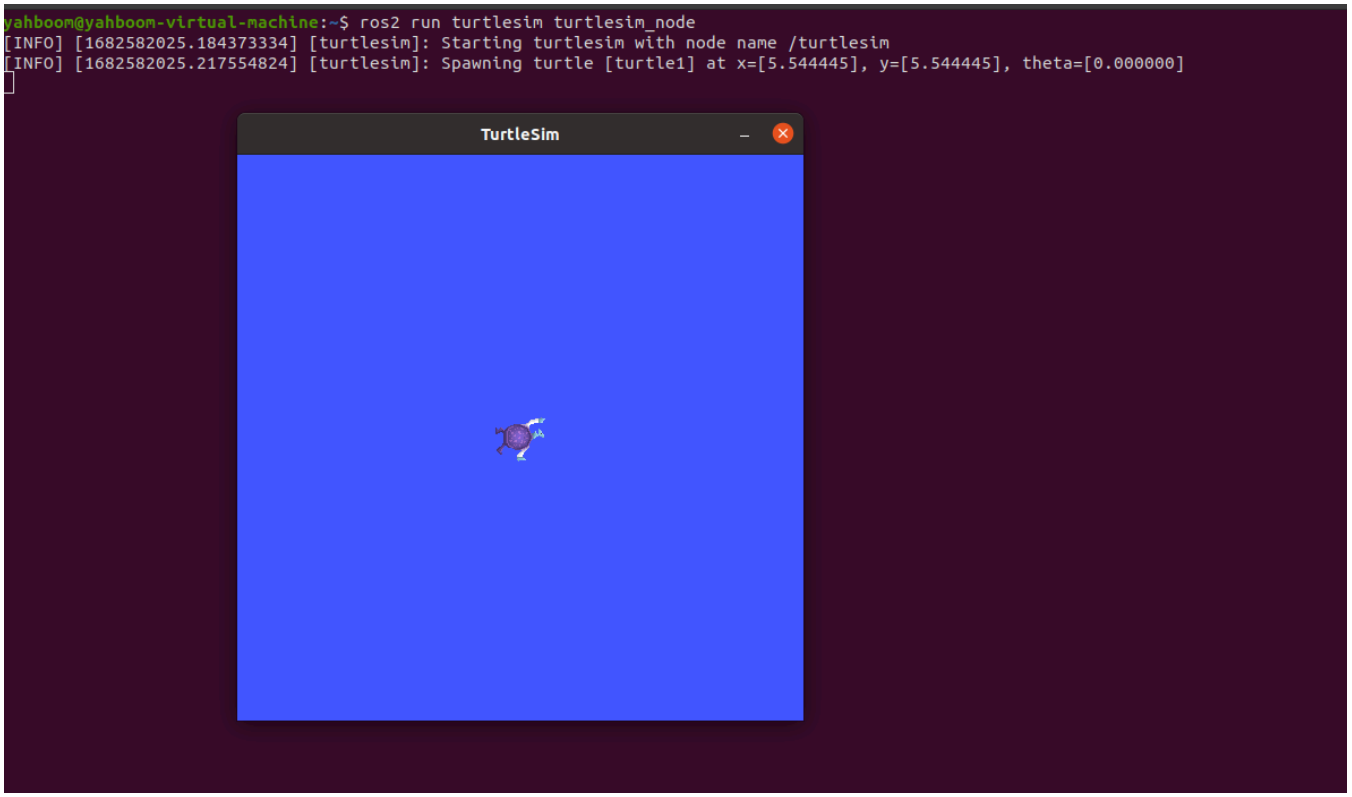
2. Node Run ros2 run

Function: run the node program of a function package

Format:

```
ros2 run pkg_name node_name
```

- pkg_name: the name of the function package
- node_name: the name of the executable program



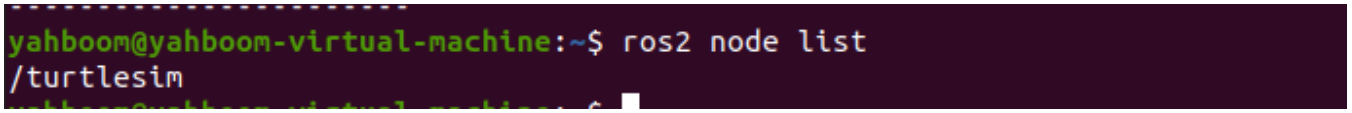
3. Node-Related Tools ros2 node

3.1. ros2 node list

Function: list the names of all nodes in the current domain

Format:

```
ros2 node list
```



3.2. ros2 node info

Function: view detailed node information, including subscribed and published messages, started services and actions, etc.

Format:

```
ros2 node info node_name
```

- node_name: the name of the node to be viewed

```
yahboom@yahboom-virtual-machine:~$ ros2 node info /turtlesim
/turtlesim
  Subscribers:
    /parameter_events: rcl_interfaces/msg/ParameterEvent
    /turtle1/cmd_vel: geometry_msgs/msg/Twist
  Publishers:
    /parameter_events: rcl_interfaces/msg/ParameterEvent
    /rosout: rcl_interfaces/msg/Log
    /turtle1/color_sensor: turtlesim/msg/Color
    /turtle1/pose: turtlesim/msg/Pose
  Service Servers:
    /clear: std_srvs/srv/Empty
    /kill: turtlesim/srv/Kill
    /reset: std_srvs/srv/Empty
    /spawn: turtlesim/srv/Spawn
    /turtle1/set_pen: turtlesim/srv/SetPen
    /turtle1/teleport_absolute: turtlesim/srv/TeleportAbsolute
    /turtle1/teleport_relative: turtlesim/srv/TeleportRelative
    /turtlesim/describe_parameters: rcl_interfaces/srv/DescribeParameters
    /turtlesim/get_parameter_types: rcl_interfaces/srv/GetParameterTypes
    /turtlesim/get_parameters: rcl_interfaces/srv/GetParameters
    /turtlesim/list_parameters: rcl_interfaces/srv/ListParameters
    /turtlesim/set_parameters: rcl_interfaces/srv/SetParameters
    /turtlesim/set_parameters_atomically: rcl_interfaces/srv/SetParametersAtomically
  Service Clients:

  Action Servers:
    /turtle1/rotate_absolute: turtlesim/action/RotateAbsolute
  Action Clients:
```

4. Topic-Related Tools ros2 topic

4.1. ros2 topic list

Function: list all the topics in the current domain

Format:

```
ros2 topic list
```

```
yahboom@yahboom-virtual-machine:~$ ros2 topic list
/parameter_events
/rosout
/turtle1/cmd_vel
/turtle1/color_sensor
/turtle1/pose
```

4.2. ros2 topic info

Function: display the message type of a topic and the number of subscribers/publishers

Format:

```
ros2 topic info topic_name
```

- topic_name: the name of the topic to be queried

```
yahboom@yahboom-virtual-machine:~$ ros2 topic info /turtle1/cmd_vel
Type: geometry_msgs/msg/Twist
Publisher count: 0
Subscription count: 1
```

4.3. ros2 topic type

Function: view the message type of a topic

Format:

```
ros2 topic type topic_name
```

- topic_name: the name of the topic whose type needs to be queried

```
yahboom@yahboom-virtual-machine:~$ ros2 topic type /turtle1/cmd_vel
geometry_msgs/msg/Twist
```

4.4. ros2 topic hz

Function: display the average publishing frequency of a topic

Format:

```
ros2 topic hz topic_name
```

- topic_name: the name of the topic whose frequency needs to be queried

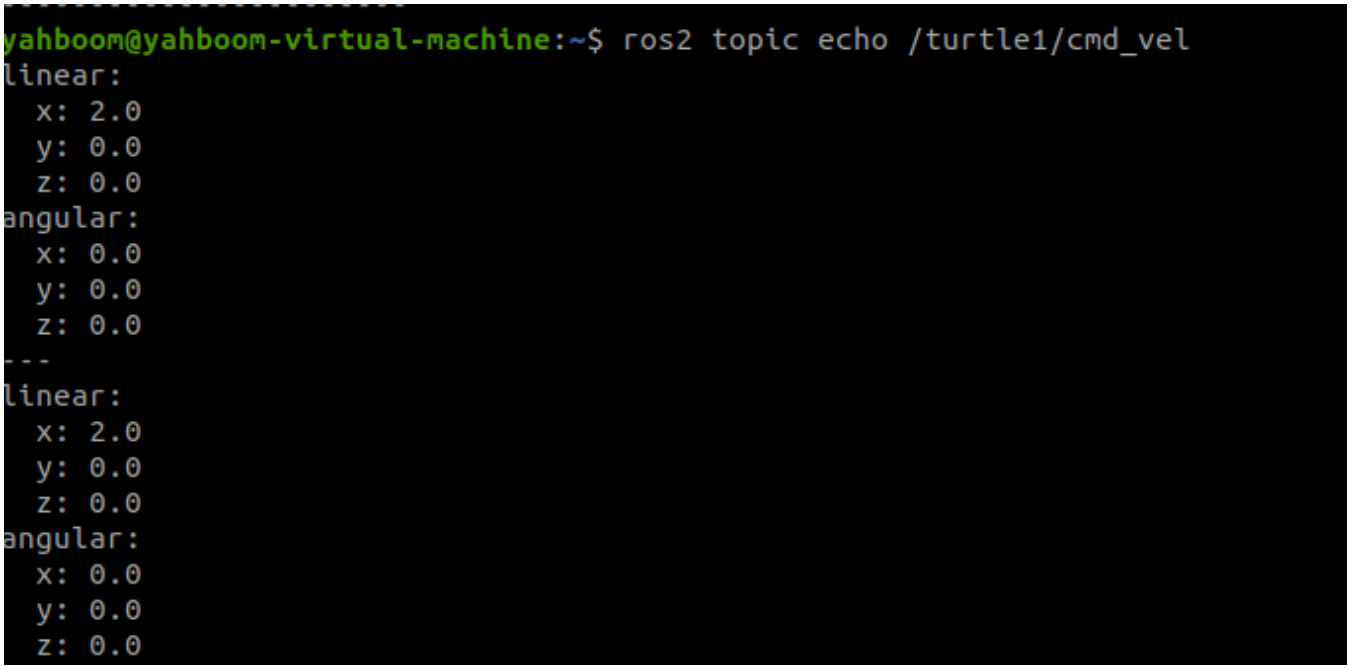
```
yahboom@yahboom-virtual-machine:~$ ros2 topic hz /turtle1/cmd_vel
average rate: 2.532
        min: 0.002s max: 6.513s std dev: 1.44588s window: 19
average rate: 4.026
        min: 0.002s max: 6.513s std dev: 1.06690s window: 36
average rate: 4.613
        min: 0.002s max: 6.513s std dev: 0.93960s window: 47
average rate: 5.803
        min: 0.002s max: 6.513s std dev: 0.80420s window: 65
average rate: 5.961
        min: 0.002s max: 6.513s std dev: 0.75605s window: 74
average rate: 5.991
        min: 0.002s max: 6.513s std dev: 0.72046s window: 82
average rate: 5.755
        min: 0.002s max: 6.513s std dev: 0.70435s window: 86
average rate: 5.568
        min: 0.002s max: 6.513s std dev: 0.68547s window: 91
average rate: 5.419
        min: 0.002s max: 6.513s std dev: 0.67609s window: 94
```

4.5. ros2 topic echo

Function: print topic messages in the terminal, similar to a subscriber

Format: ros2 topic echo topic_name

- topic_name: the name of the topic whose messages need to be printed



4.6. ros2 topic pub

Function: publish messages on a specified topic in the terminal

Format:

```
ros2 topic pub topic_name message_type message_content
```

- topic_name: the name of the topic on which to publish the message
- message_type: the data type of the topic
- message_content: the message content

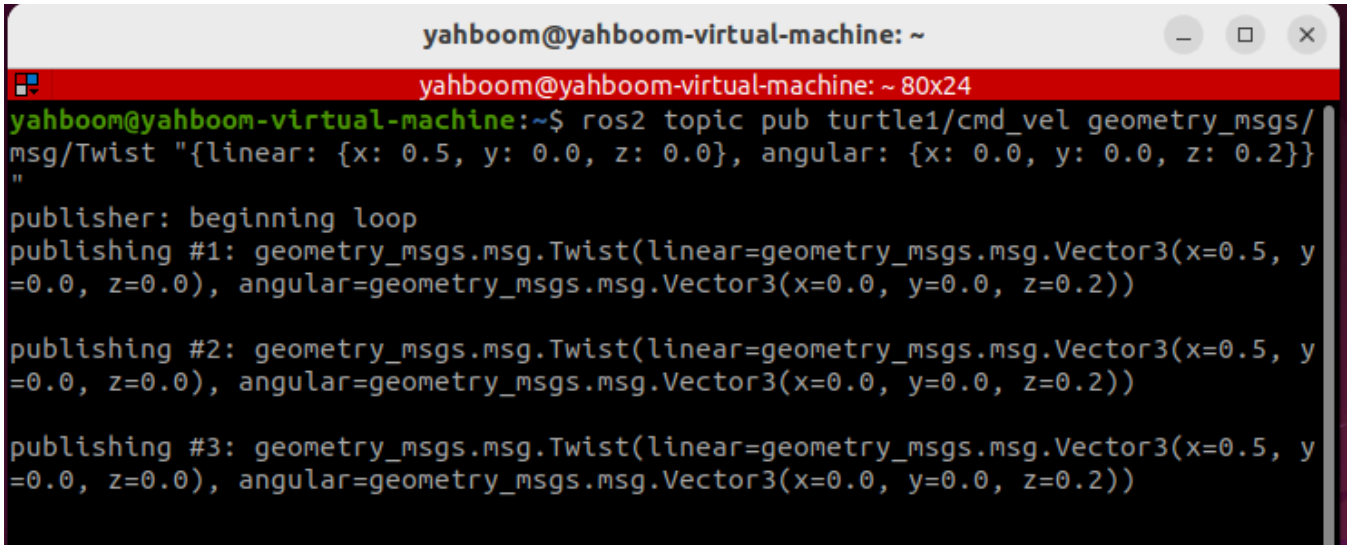
By default it publishes in a loop at 1Hz. The following parameters can be set:

- Parameter -1 publishes only once, ros2 topic pub -1 topic_name message_type message_content
- Parameter -t count publishes in a loop count times and then ends, ros2 topic pub -t count topic_name message_type message_content
- Parameter -r count publishes in a loop at a frequency of count Hz, ros2 topic pub -r count topic_name message_type message_content

Example:

- Publish a speed command through the command line
- Note that there should be a space after each colon, otherwise a format error will be reported

```
ros2 topic pub turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.5, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.2}}"
```



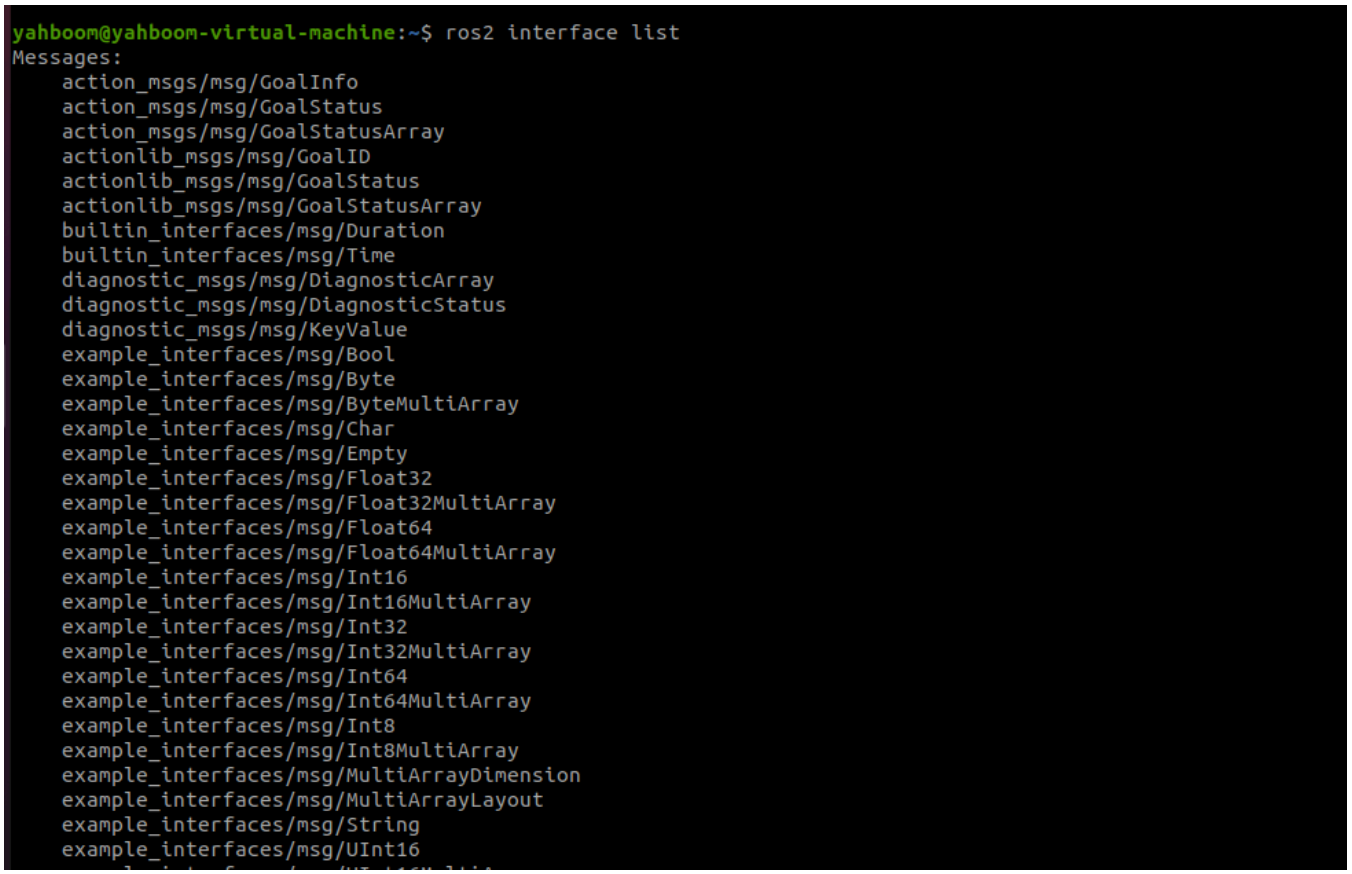
5. Interface-Related Tools ros2 interface

5.1. ros2 interface list

Function: list all the interfaces in the current system, including topics, services, and actions.

Format:

```
ros2 interface list
```



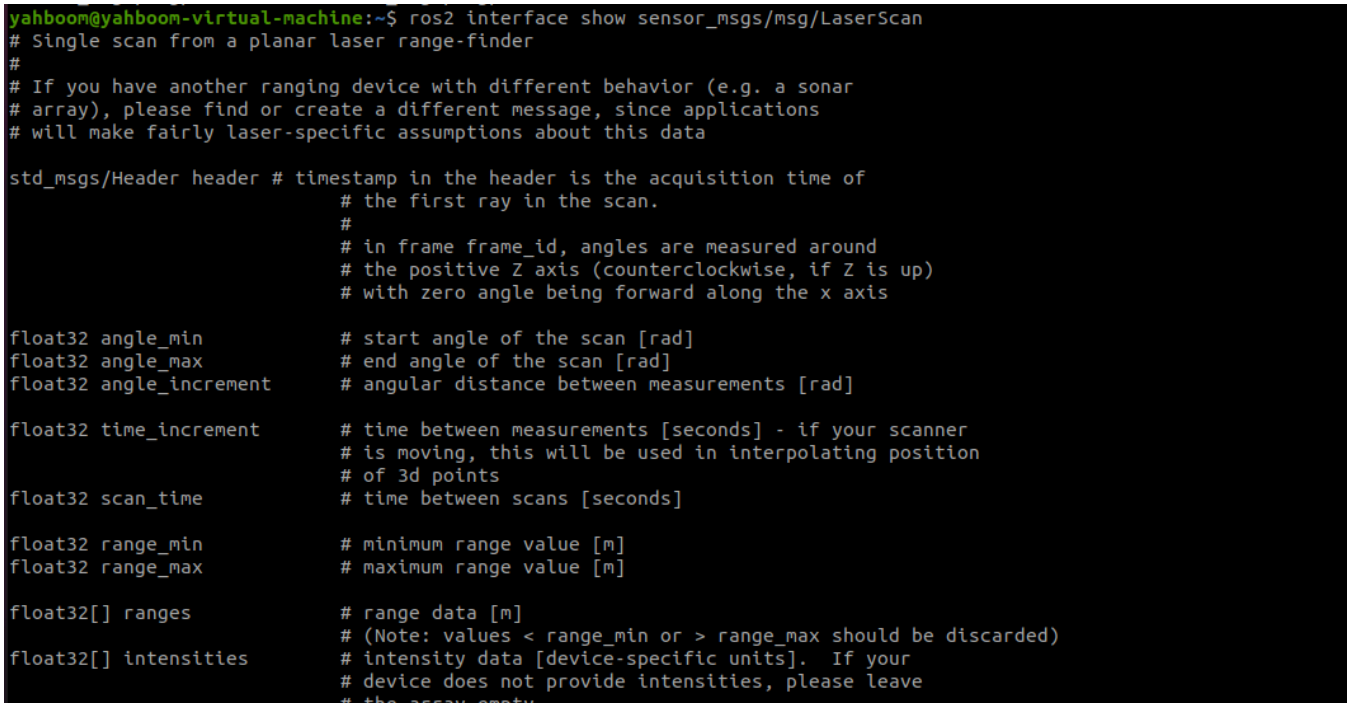
5.2. ros2 interface show

Function: display the detailed content of a specified interface

Format:

```
ros2 interface show interface_name
```

- interface_name: the name of the interface whose content needs to be displayed



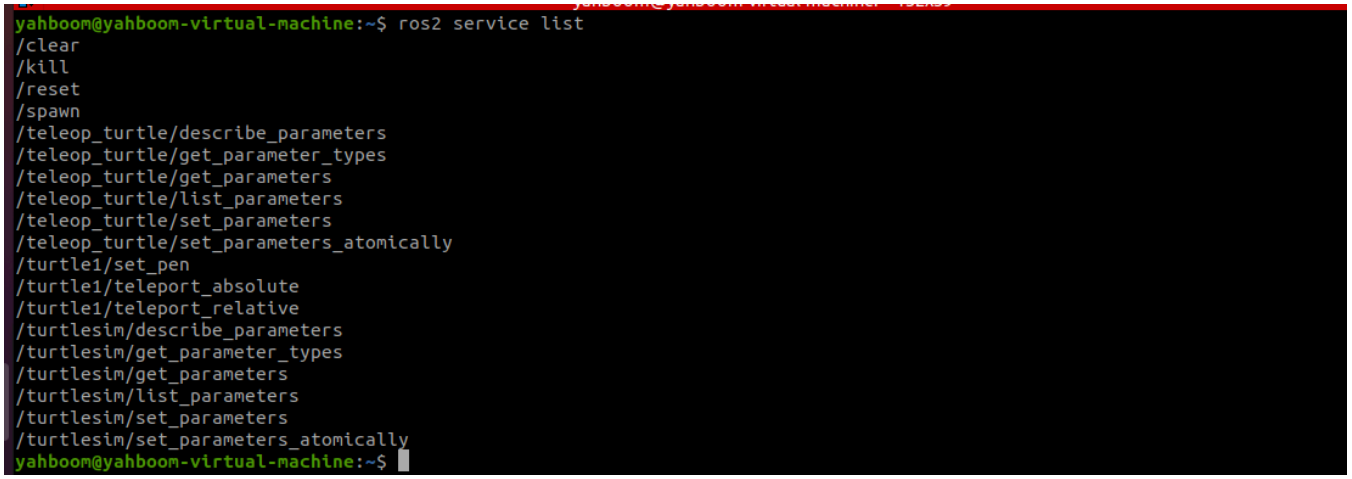
6. Service-Related Tools ros2 service

6.1. ros2 service list

Function: list all the services in the current domain

Format:

```
ros2 interface show interface_name
```



6.2. ros2 service call

Function: call a specified service

Format:

```
ros2 interface call service_name service_Type arguments
```

- service_name: the service to be called
- service_Type: the service data type
- arguments: the parameters needed to provide the service

For example, call the turtle spawn service

```
ros2 service call /spawn turtlesim/srv/Spawn "{x: 2, y: 2, theta: 0.2, name: 'turtle10'}"
```

